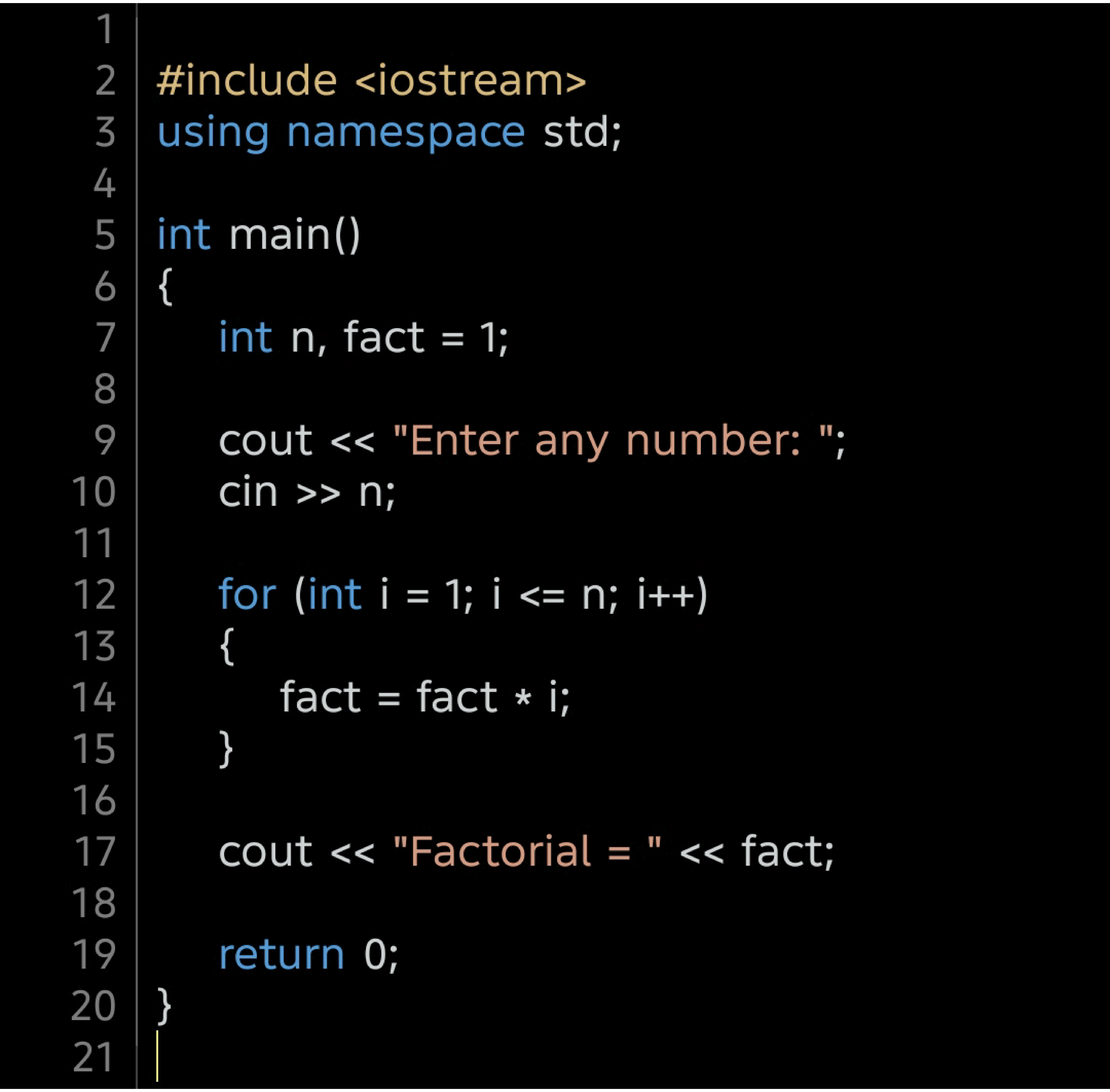
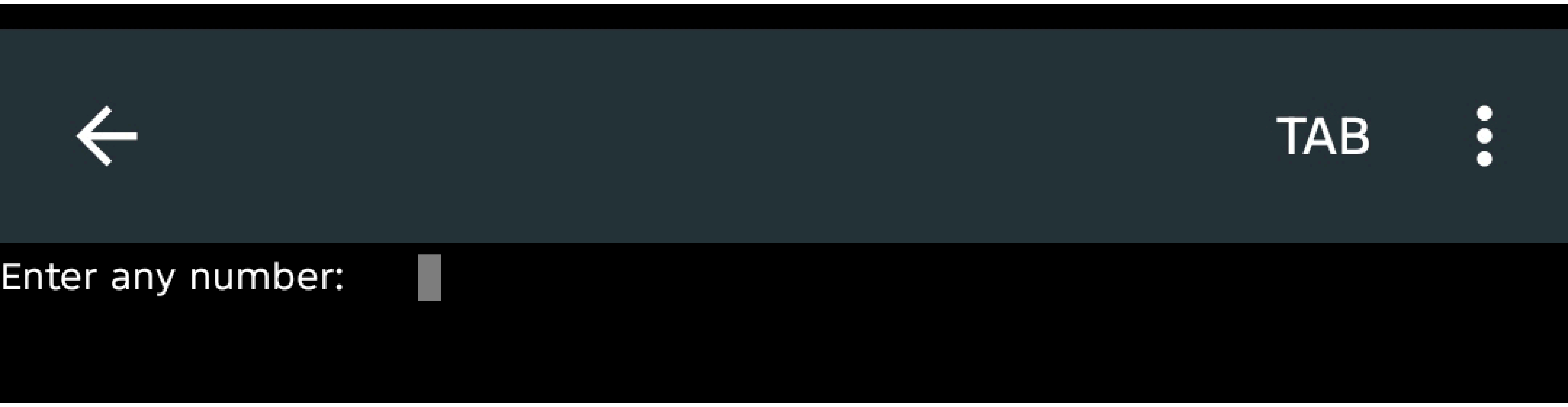


```cpp

#include <iostream>
using namespace std;

int main()
{
    int n, fact = 1;

    cout << "Enter any number: ";
    cin >> n;

    for (int i = 1; i <= n; i++)
    {
        fact = fact * i;
    }

    cout << "Factorial = " << fact;

    return 0;
}

```



```
Enter any number:
```

Date:

Lab Task: 05

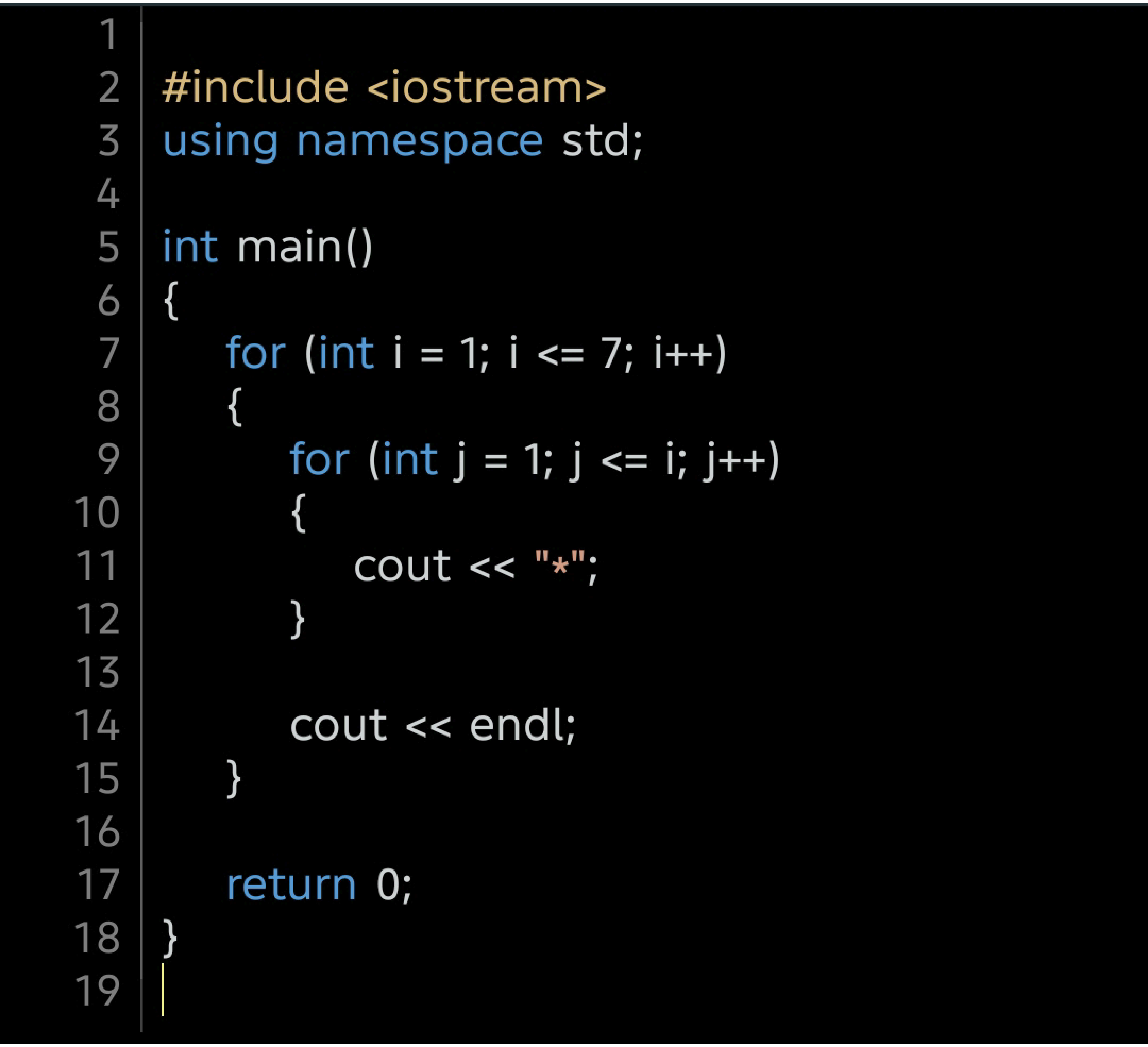
(1)

C++ program to print Table of a
number.

```
#include<iostream>
 using namespace std;
  int main()
  {
  int num;
  Cout<<"Enter a number:";
  Cin>> num;
 for(int i=1; i<=10; i++)
   {
   Cout<< <<" x" << i <<"="
          << num*i<<
  endl;
   }
return 0;
}
```

Output:

```
Enter a numbe: 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```



```

#include <iostream>
using namespace std;

int main()
{
    for (int i = 1; i <= 7; i++)
    {
        for (int j = 1; j <= i; j++)
        {
            cout << "*";
        }

        cout << endl;
    }

    return 0;
}

```

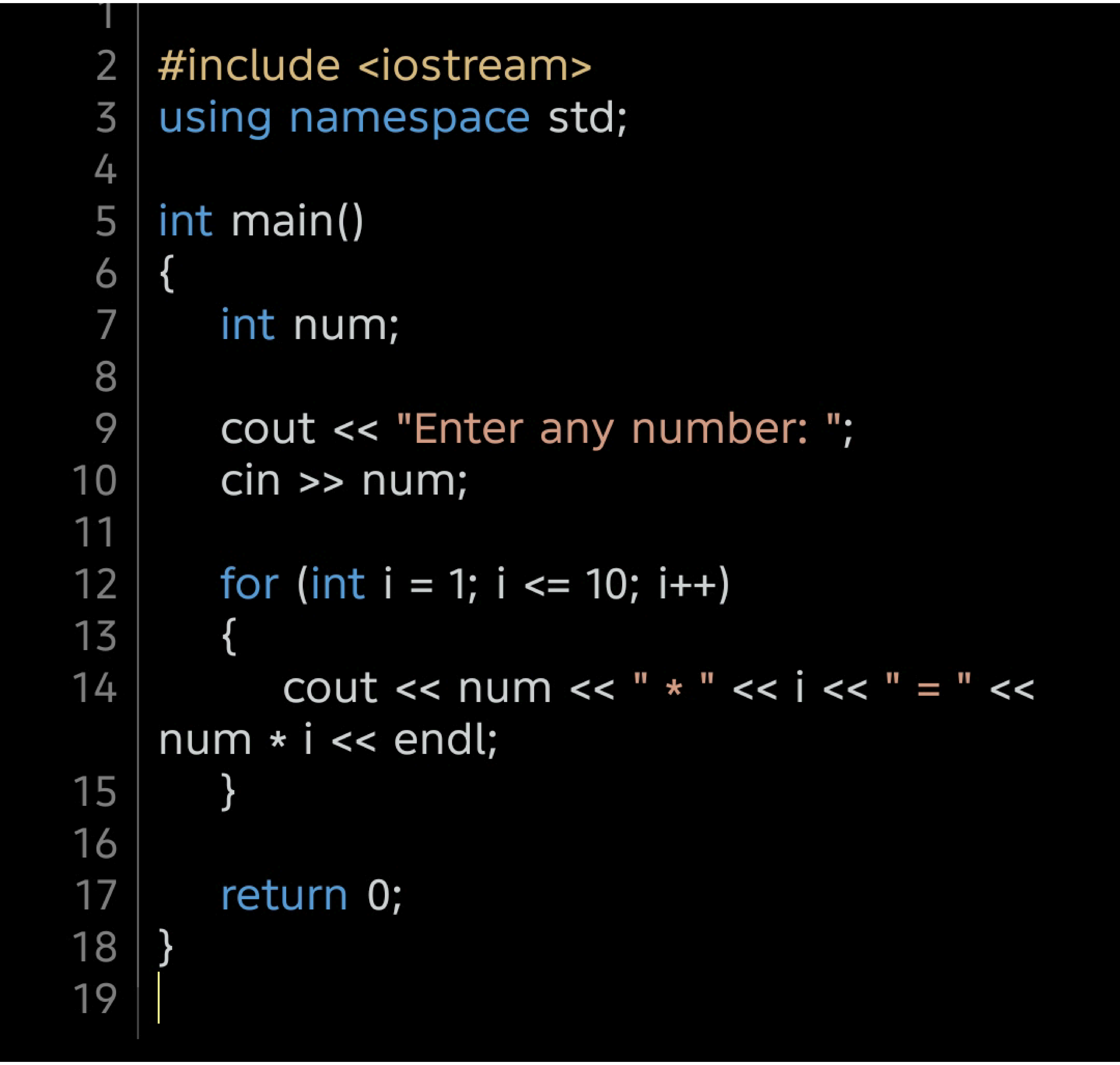
(2)

Find the power of Any Number Using a loop.

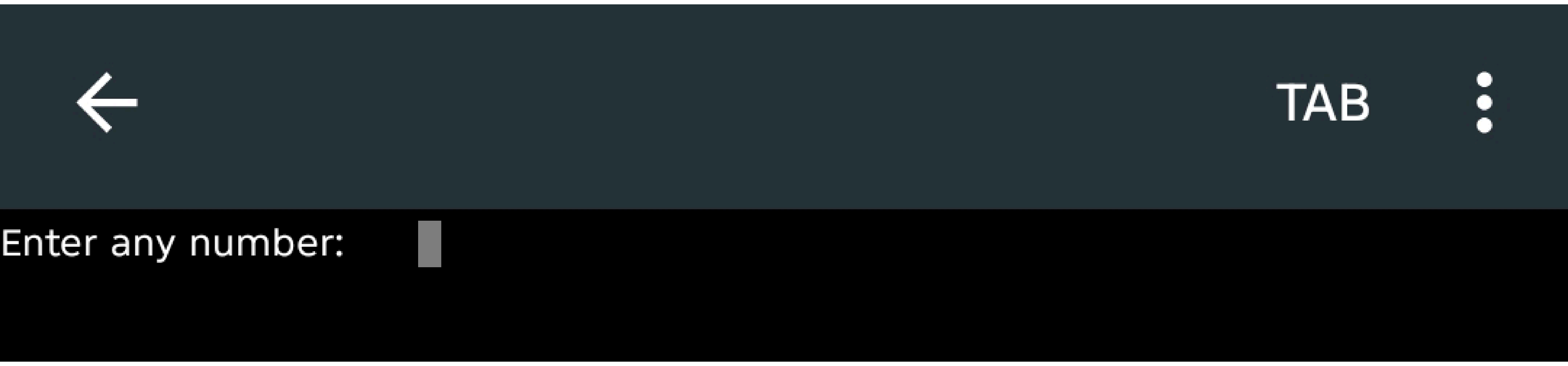
```
#include <iostream>
 Using namespace std;
 int main()
{
 int base, exponent;
 int power = 1;
 Cout << "Enter the base: ";
 Cin >> base;
Cout << "Enter the exponent: ";
cin >> exponent:
 if( exponent < 0)
 {
Cout << "please enter a non-negative
exponent.";
 }
else
{
for(int i=1: i<= exponent; i++)
  {
  power= power* base;
  }
Cout<< "power = "<< power;
 }
retuono:
}
```

Output:

```
    Enter the base: 2
    Enter the exponent: 4
  power = 16
```



```cpp
#include <iostream>
using namespace std;

int main()
{
    int num;

    cout << "Enter any number: ";
    cin >> num;

    for (int i = 1; i <= 10; i++)
    {
        cout << num << " * " << i << " = " << num * i << endl;
    }

    return 0;
}
```



```
Enter any number:
```

(3)

Sum of Natural using loop.

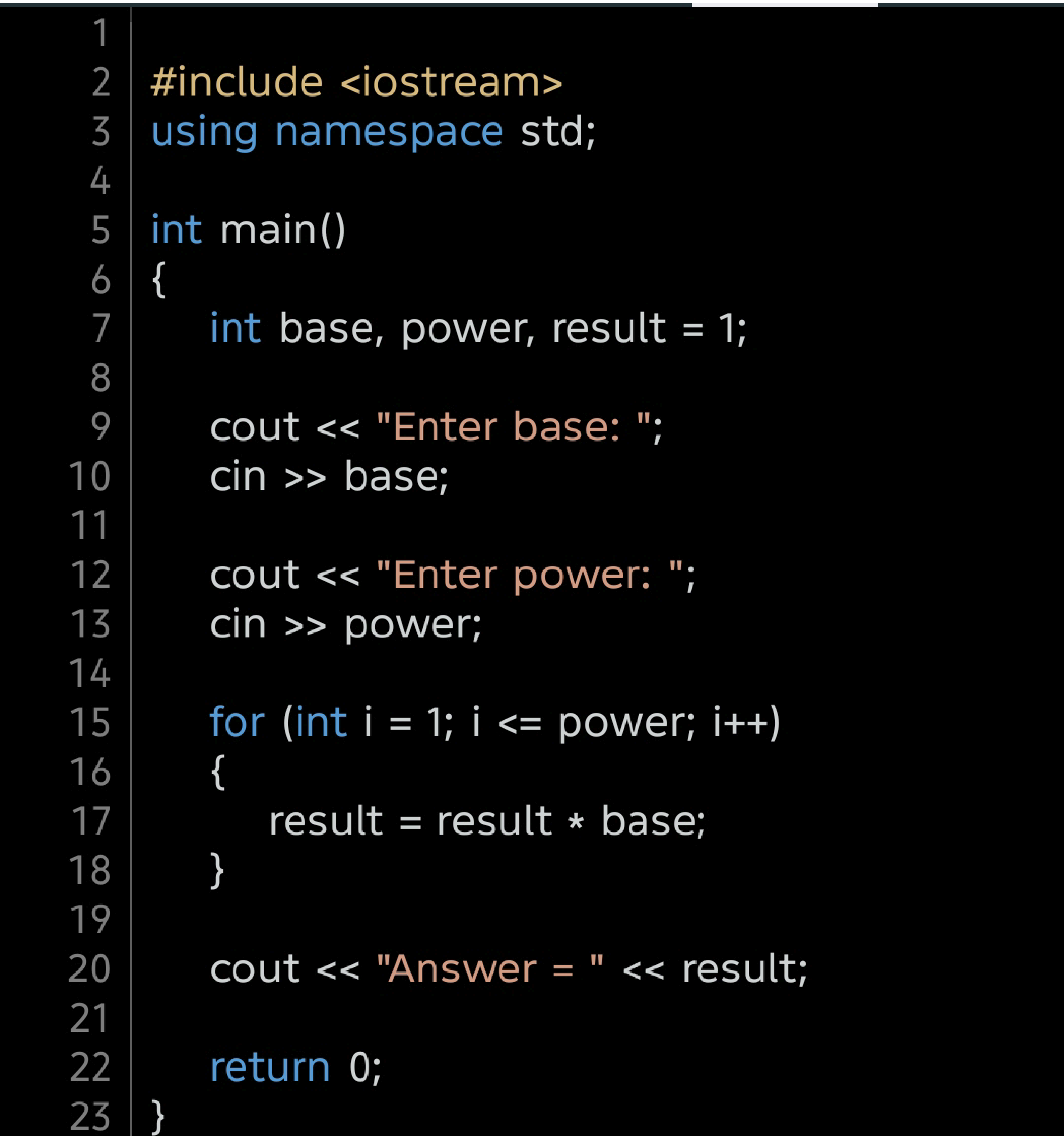
```
#include<iostream>
using namespace std;
int main()
{
int n, sum=0;
Cout<< Enter positive number:";
 cin>>n;
 if(int = 1)
 Cout<<"please enter a positive num.";
}
else
{
for(int i=1; i<=n; i++)
{
sum = sum i;
 }

Cout<<"Sum = "<< sum;
```

Date:

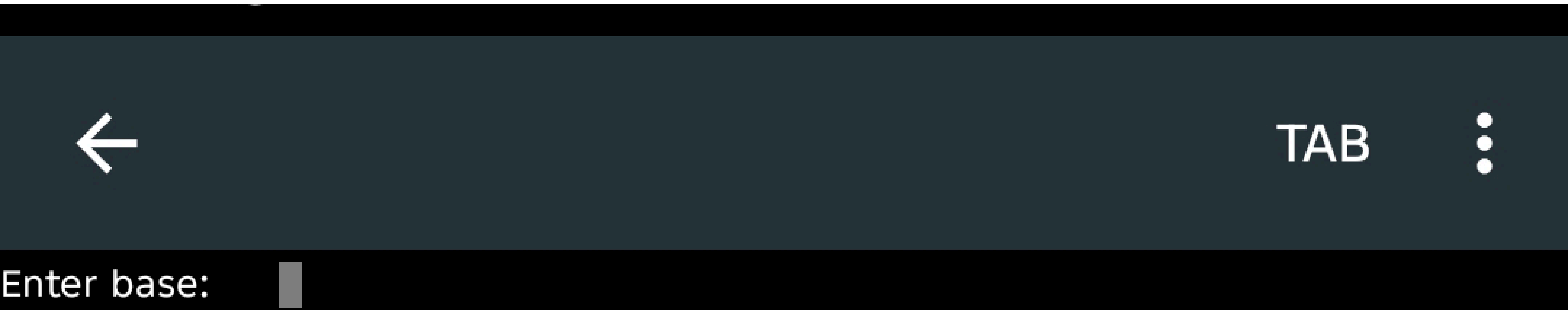
```
}
return 0;
}
```

Output:

```
    Enter a positive number: 5
  Sum = 15
```



```cpp

#include <iostream>
using namespace std;

int main()
{
    int base, power, result = 1;

    cout << "Enter base: ";
    cin >> base;

    cout << "Enter power: ";
    cin >> power;

    for (int i = 1; i <= power; i++)
    {
        result = result * base;
    }

    cout << "Answer = " << result;

    return 0;
}
```



```
Enter base:
```

Date:__________

(4)

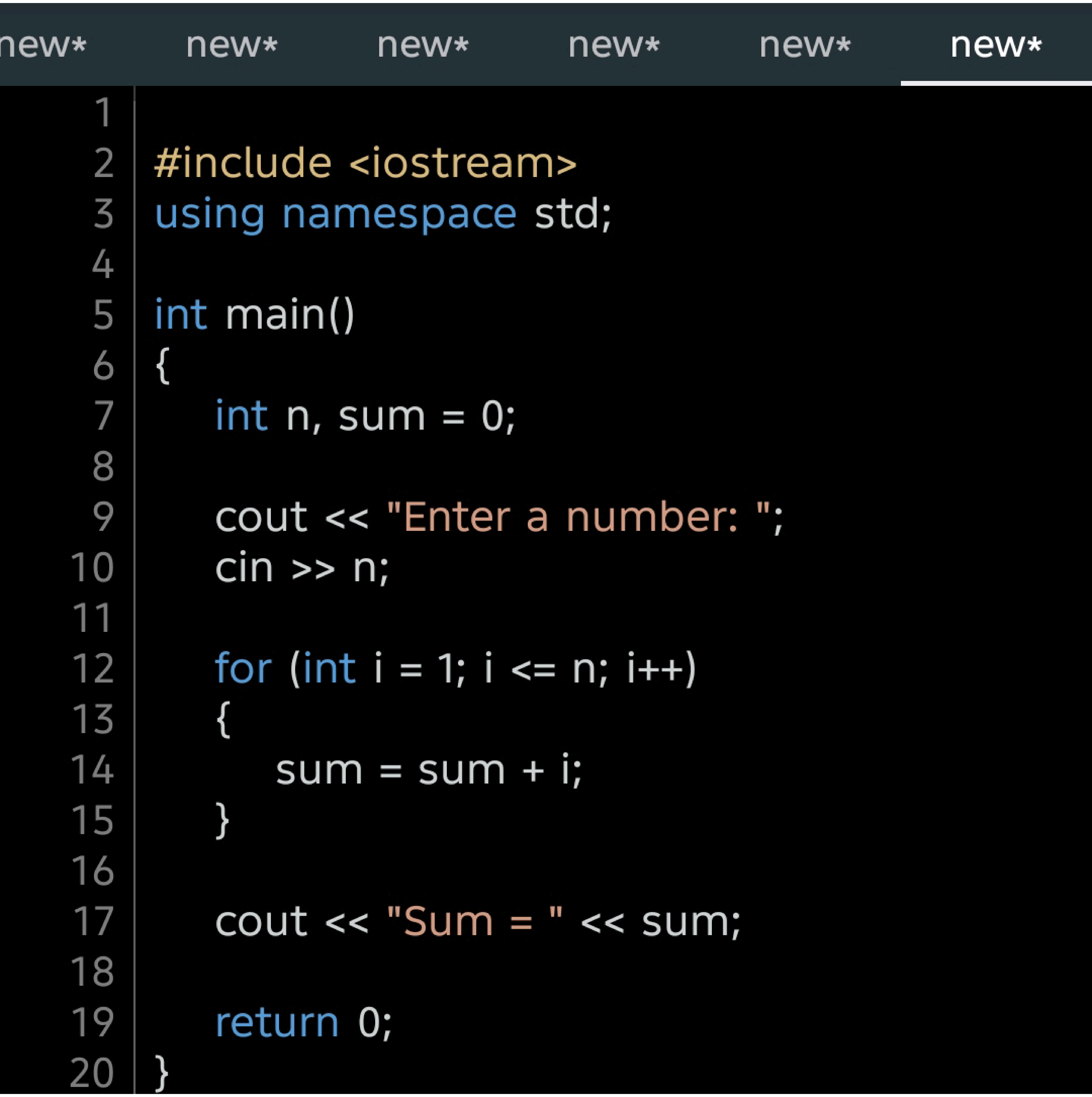
C++ program to find Sum of digits of a number.

```
#include <iostream>
using namespace std;
int main()
{
int num, digit, sum = 0;
Cout<<"Enter a number: ";
 Cin>> num;
 if (num<0)
 {
  num = num:
 }
while (num>0)
}
digit = num% 10:
Sum = Sum + digit:
 num = num / 10;
```

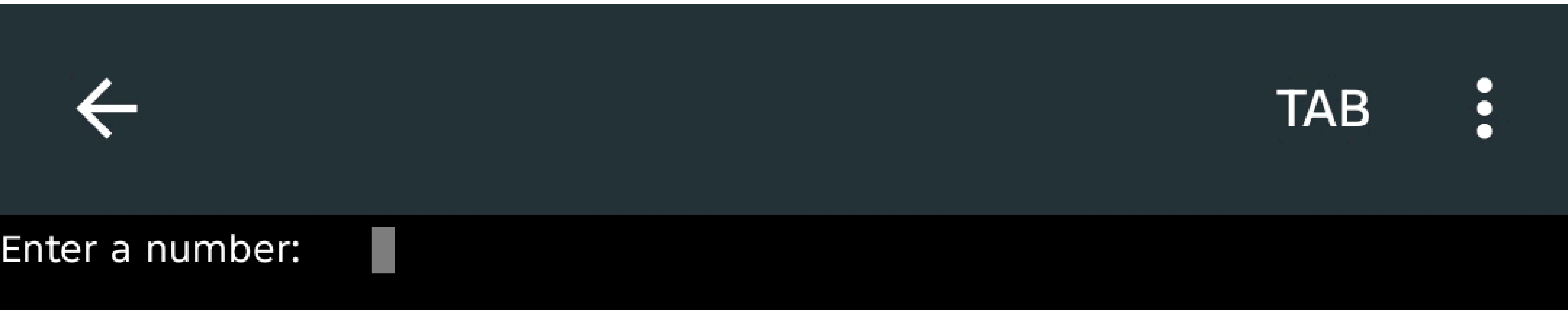
```
}
Cout<<"Sum of digits ="<<sum;
return 0;
}
```

Output:

```
    Enter a number: 358
    Sum of digits = 16
```



```

#include <iostream>
using namespace std;

int main()
{
    int n, sum = 0;

    cout << "Enter a number: ";
    cin >> n;

    for (int i = 1; i <= n; i++)
    {
        sum = sum + i;
    }

    cout << "Sum = " << sum;

    return 0;
}
```



```
Enter a number:
```

Date:

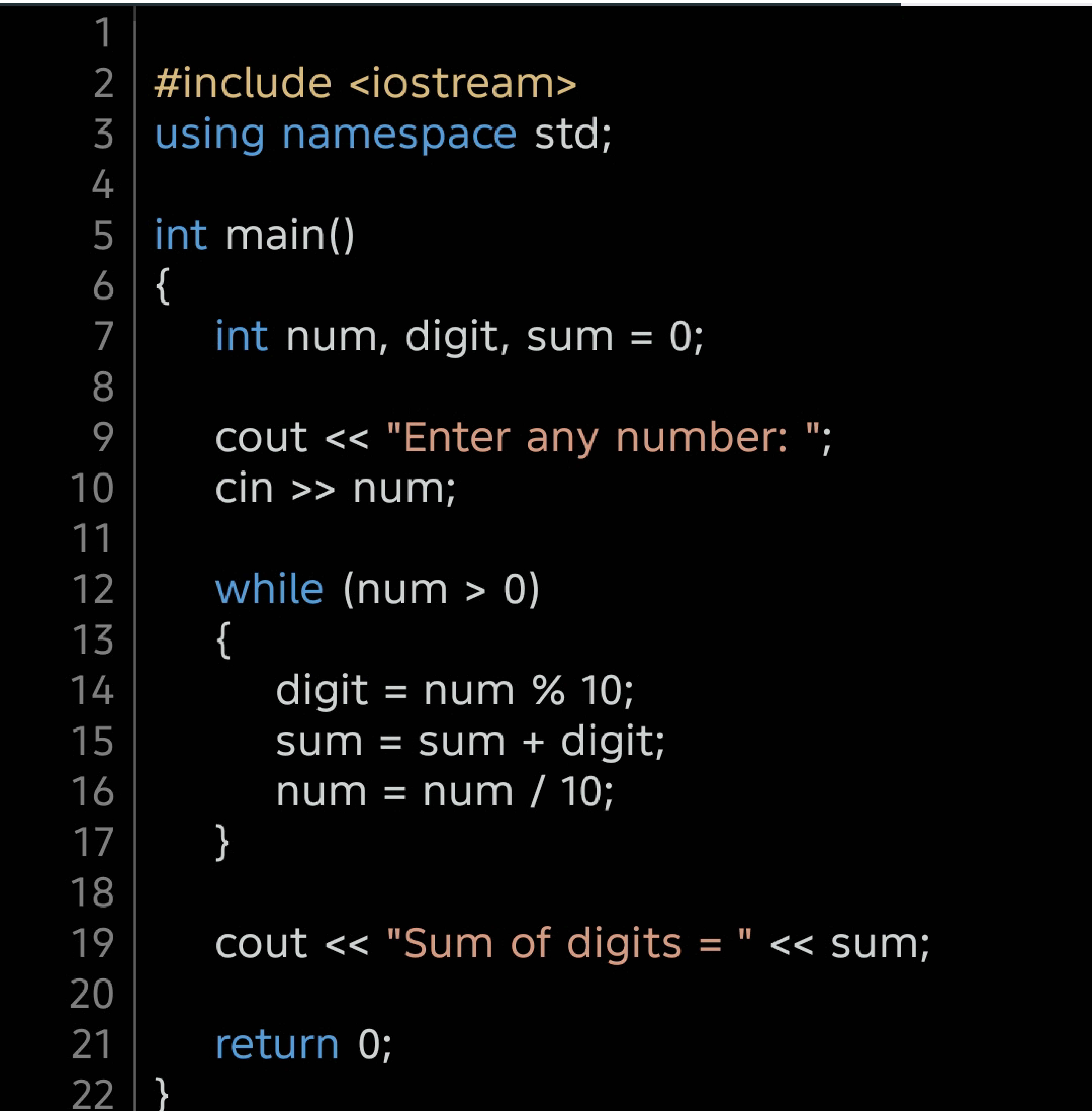
(5)

program to generate Factorial of a certain number

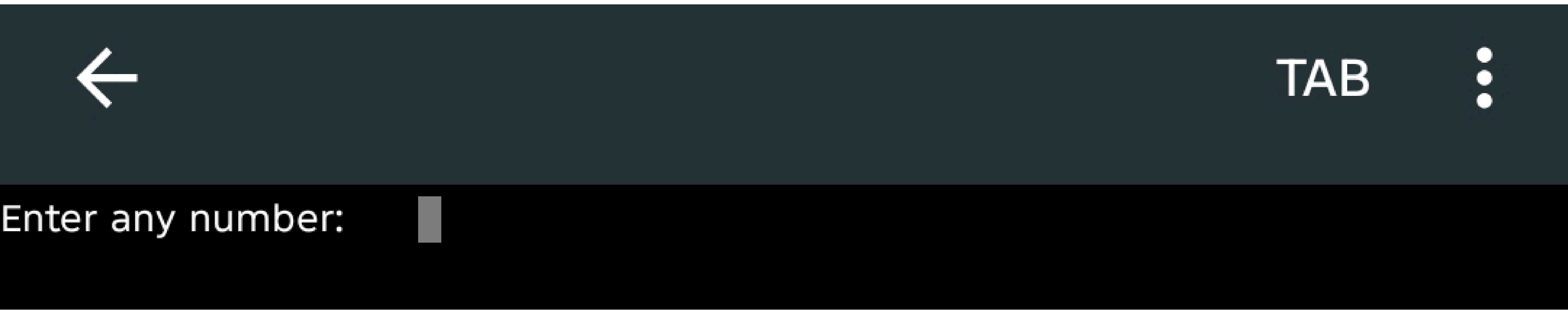
```
#include<iostream>
Using namespace std;
int main()
{
  int n, fact =1;
  Cout<<"Enter any number:";
  Cin>>n;
  for(int i=1; i<=n; i++)
  {
   fact = fact*i
   Cout<<"factorial="<<fact;
   return 0;
```

Output.

```
Enter any number: 4
 factorial = 24
```



```cpp

#include <iostream>
using namespace std;

int main()
{
    int num, digit, sum = 0;

    cout << "Enter any number: ";
    cin >> num;

    while (num > 0)
    {
        digit = num % 10;
        sum = sum + digit;
        num = num / 10;
    }

    cout << "Sum of digits = " << sum;

    return 0;
}
```



```
Enter any number:
```

Date:

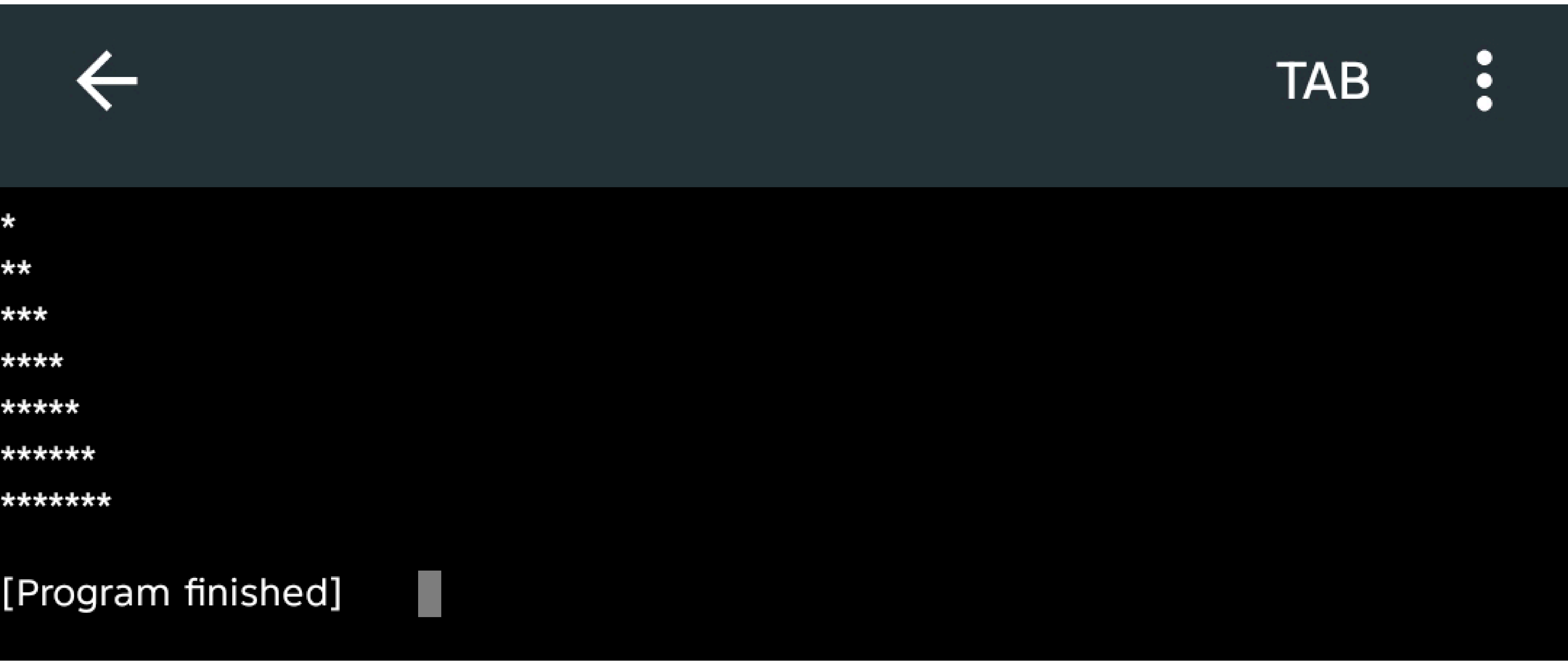
(6)

print the following pattern

```
#include<iostream>
Using namespace std;
int main()
{
for (int i=1; i<=7; i++)
 {
for (int j = 1; j <= i; j++)
{
 cout << "*";
 }
 cout << endl;
 }
 return 0;
 }
```

Output:

```
   C++
for (int i = 1; i<=7; i++)
```



```
*
**
***
****
*****
******
*******

[Program finished]
```